# Quadric

Joseph Sullivan

June 2025

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# 1 Setup

Let $E$ be a finite dimensional $\mathbb{C}$ vector space, $\langle -, - \rangle$ a nondegenerate symmetric bilinear form on $E$, and $Q$ the associated quadratic form (we will identify the equation and the projective variety cut out of $\mathbb{P}E$). We will write $E$ to mean the quadratic space, a pair of a vector space and a quadratic form. These notes will describe how to get a full strong exceptional collection on the quadric

$$Q \subseteq \mathbb{P}E = \operatorname{Proj} \operatorname{Sym}^\bullet E^*.$$

# 2 Quadratic Algebras

Corresponding to $Q$ is the homogeneous coordinate ring

$$B = B(E) = \operatorname{Sym}^\bullet E^*/(Q),$$

which is a quadratic algebra, so in our study we will also study the quadratic dual, which we will refer to as a *graded* Clifford algebra

$$A = A(E) = T^\bullet E/(Q(\eta)\xi^2 + Q(\xi)\eta^2).$$

We can also be conveniently describe it by adding a degree 2 symbol $h$ to stand for (a multiple of) any nonzero $\xi^2$, i.e.,

$$A \cong T(E \oplus h)/(Q(\xi)h - \xi^2, \xi h - h\xi).$$

(The relation $Q(\xi)h = \xi^2$ is also equivalent to $2\langle \xi, \eta \rangle h = \xi\eta + \eta\xi$ at least when the characteristic of the field is not 2).

This really generalizes the story of projective space, where we study the rings $\mathrm{Sym}^\bullet E^*$ and $\bigwedge^\bullet E$ which are again quadratic duals.

**Proposition 2.1.** *$A$ and $B$ are Koszul rings.*

*Proof.* Check that the associated Koszul complex

$$\cdots \to A_2^* \otimes B \to A_1^* \otimes B \to A_0^* \otimes B \to \mathbb{C} \to 0$$

is exact. this is apparently a well-known thing, this is the "Tate resolution"? ☐

We will use this Koszul/quadratic duality in particular in how it relates modules and linear projective complexes, as in the following proposition.

**Proposition 2.2.** *Let $A, B$ be quadratic dual algebras. Then,*

$$\mathrm{Gr}A\text{-Mod} \simeq \mathcal{LC}_B,$$

*where $\mathcal{LC}_B$ is the category of linear projective complexes over $B$.*

*Proof.* We define the functors forming the inverse equivalence. Let $M$ be a graded $B$-module. We define a complex $\mathcal{L}(M)$

$$\cdots \to M_k \otimes A \to M_{k+1} \otimes A \to \cdots,$$

where the differential is the data of a map

$$M_k \to M_{k+1} \otimes A_1.$$

We get this differential by considering the multiplication map

$$M_k \otimes B_1 \to M_{k+1},$$

which we tensor with $A_1 = B_1^*$ and precompose with the map

$$M_k \otimes k \to M_k \otimes A_1 \otimes B_1$$

sending $1 \in k$ to $\mathrm{eval} \in A_1 \otimes B_1$.

Going backwards, we take a linear projective complex $(M^k, d)$ and define a $B$-module $M$ by first defining the $k$-vector space

$$M = \bigoplus M^k,$$

and to define the $B$-module structure, we just need to give $B_1$ multiplication (satisfying the relations). For $b \in B_1$ and $m \in M_k$, we define

$$b \cdot m := \mathrm{eval}_b(d(1 \otimes m))$$

since $d(1 \otimes m) \in A_1 \otimes M_{k+1}$. ☐

In particular, the Koszul complex for resolving $\mathbb{C}$ as a $B$-module is $\mathcal{L}(A^*)$.

## 3 Resolution of the Diagonal

Now, our strategy to find a full strong exceptional collection for $Q$ is to repeat Beilinson's strategy for $\mathbb{P}E$. We recall that Beilinson's resolution of the diagonal is a complex of the form

$$\cdots \to \Omega^i(i) \boxtimes \mathcal{O}(-i) \to \Omega^{i-1}(i-1) \boxtimes \mathcal{O}(-i+1) \to \cdots,$$

and in fact we can recognize the $\Omega^i(i)$ as twists of kernels in the Koszul complex for $\mathbb{P}E$, i.e., the Euler sequence implies

$$\Omega^i \cong \ker\left(\bigwedge^i E \otimes_{\mathbb{C}} \mathcal{O}(-i) \to \bigwedge^{i-1} E \otimes_{\mathbb{C}} \mathcal{O}(-i+1)\right).$$

In analogy, we might expect

$$\Psi_i := \ker\left(A_i^* \otimes_{\mathbb{C}} \mathcal{O} \to A_{i-1}^* \otimes_{\mathbb{C}} \mathcal{O}(1)\right)$$

to play an important role.

In Beilinson's argument, the resolution of the diagonal was the Koszul complex for $\Delta \subseteq \mathbb{P}E \times \mathbb{P}E$ being a local complete intersection cut out by (the dual of) the evaluation section of the bundle $\Omega^1(1) \boxtimes \mathcal{O}(-1)$.

Likewise, for Kapranov's argument we will use a Koszul complex in the product, but this is in the sense of quadratic/Koszul duality, not the traditional commutative Koszul complex—this is because it just isn't true that $\Delta_Q$ is a local complete intersection (the bundle $\Omega^1(1)|_Q$ is too high rank). So, we will need a more sophisticated resolution, which in our case will be infinite.

**Proposition 3.1.** There is a complex of sheaves on $Q \times Q$

$$\cdots \to \Psi_2 \boxtimes \mathcal{O}(-2) \to \Psi_1 \boxtimes \mathcal{O}(-1) \to \mathcal{O}_{Q\times Q} \to 0$$

which is a resolution of the diagonal.

*Proof.* We will write down this complex by working instead with graded modules. Equipping $Q \times Q$ with the very ample bundle $\mathcal{O}_{Q\times Q}(1,1)$, we get the section ring

$$B^2 = \bigoplus_i B_i \otimes B_i.$$

Now, before we lift $\Psi_j \boxtimes \mathcal{O}(-j)$ to a graded $B^2$-module, we replace it by the quasi-isomorphic complex

$$A_j^* \otimes_{\mathbb{C}} \mathcal{O} \to A_{j-1}^* \otimes_{\mathbb{C}} \mathcal{O}(1) \to \cdots \to A_0^* \otimes_{\mathbb{C}} \mathcal{O}(j),$$

which, after applying $\mathcal{O}(-j) \boxtimes (-)$, lifts to the graded complex of $B^2$-modules

$$\bigoplus_i B_{i-j} \otimes A_j^* \otimes B_i \to \bigoplus_i B_{i-j} \otimes A_{j-1}^* \otimes B_{i+1} \to \cdots \to \bigoplus_i B_{i-j} A_0^* \otimes B_{i+j}.$$

So, to specify our desired complex

$$\cdots \to \Psi_2 \boxtimes \mathcal{O}(-2) \to \Psi_1 \boxtimes \mathcal{O}(-1) \to \mathcal{O}_{Q\times Q},$$

we can specify a double complex of graded $B^2$ modules lifting these bundles, of the shape

$$\begin{tikzcd}
\cdots & & & \\
\cdots \arrow[u] \arrow[r] & \bigoplus_i B_{i-2} \otimes B_{i+2} & & \\
\cdots \arrow[u] \arrow[r] & \bigoplus_i B_{i-2} \otimes A_1^* \otimes B_{i+1} \arrow[u] \arrow[r] & \bigoplus_i B_{i-1} \otimes B_{i+1} & \\
\cdots \arrow[u] \arrow[r] & \bigoplus B_{i-2} \otimes A_2^* \otimes B_i \arrow[u] \arrow[r] & \bigoplus_i B_{i-1} \otimes A_1^* \otimes B_i \arrow[u] \arrow[r] & \bigoplus_i B_i \otimes B_i
\end{tikzcd}$$

To specify this double complex, we just declare for a fixed $i$, the rows/columns are the Koszul complex tensored with some copy of $B_i$.

Now, we want to compute the homology of the total complex of this double complex, and see that it 0 except for at the rightmost index, where it is $\bigoplus_i B_{2i}$, which is a graded $B^2$-module corresponding to $\mathcal{O}_\Delta$ (just compute $\Gamma_*(\mathcal{O}_\Delta)$).

To compute the homology, we fix a grading degree $i$ and compute the homology—this is easy, since the rows will be Koszul complexes which are exact, except for when there is a single term $B_{i-i} \otimes B_{2i} \cong B_{2i}$ as desired. (Some extra care is warranted—making sure the $B^2$-action is correct, but essentially this is it). $\square$

# 4 Truncating the Resolution

So, we have a resolution of the diagonal, but this isn't too interesting to us yet because it is infinite in length—we don't have for example a full exceptional collection (if we take all $\mathcal{O}(-i)$ or $\Psi_i$ we would at least get a full collection, but we cannot hope for exceptional because we have "too many" sheaves).

So, we hope that we can truncate the resolution by taking a kernel at some step, but still get a box tensor so we can repeat Beilinson's argument to get a small full collection (but we will still have to do work later to see that it is strong/exceptional). To recognize this kernel, we need to define the *spinor bundles.*

## 4.1 Spinor Bundles

There are two ways to construct the Spinor bundle(s) on a quadric that we will need, both making use of the (usual) **Clifford algebra**. Define

$$\mathrm{Cl}(E) := A(E)/(h-1)$$

which is a $\mathbb{Z}/2\mathbb{Z}$-graded algebra. This will contain as a subalgebra the Lie algebra $\mathfrak{so}(E)$, which is the Lie algebra to the algebraic group

$$\mathrm{SO}(E) := \{A \in \mathrm{GL}(E) \mid Q(A\xi) = Q(\xi) \text{ for all } \xi \in E, \det A = 1\}.$$

The point is that $\mathrm{SO}(E)$ is *not* simply connected, so its Lie algebra will have *more* representations, but in all honesty it only misses (up to) two irreps, the spin representations.

First, we give a representation theoretic description. Some ingredients here will go into alterna

Let $\mathrm{Spin}(E)$ denote the complex spin group on $E$ associated to the quadratic form $Q$

Let $\mathrm{Cl}(E) = A(E)/(h-1)$ be the usual Clifford algebra, which is a $\mathbb{Z}/2\mathbb{Z}$-graded algebra. We have two cases